

CamillaFIR Output Interpretation Guide (Extended Edition)

This guide explains how to read CamillaFIR dashboard plots and generation summaries. It includes a real example screenshot and practical interpretation guidance.

1. Dashboard Overview

The dashboard contains five main sections: Magnitude & Alignment, Phase, Group Delay, Filter Gain Curve, and A-FDW Effective Bandwidth.



2. How To Read Each Section

- Magnitude & Alignment: Predicted curve should follow target smoothly without narrow extreme boosts.
- Phase: Smooth transitions are healthy. Phase clamp indicates protective limiting.
- Group Delay: Large bass peaks indicate room modes. Narrow spikes suggest reflections.
- Filter (dB): Keep boost $\leq +3$ dB for safe headroom.
- A-FDW Bandwidth: Adaptive smoothing prevents overfitting noise.

3. Evaluating Filter Quality

- Target Match > 90%
- RMS error < 1 dB
- Boost limited to safe levels
- No excessive clamp warnings
- Smooth GD below 500 Hz

If these conditions are met, the filter is technically sound. Remember: room treatment is always more effective than excessive EQ.